

Lecture 6 Team Projects

Bond Portfolio Construction

Table of contents

1	Project 1: Build a Practical Bond Index Sample	1
2	Project 2: Rebalance Under Real-World Constraints	2
3	Project 3: Bullet, Barbell, or Ladder?	2
4	Project 4: Design an Investment-Grade Mandate	2

1 Project 1: Build a Practical Bond Index Sample

Scenario. Build a transparent sample for a fund that cannot hold every bond in its benchmark.

Objectives. Define maturity, sector, quality, and duration cells; select a feasible sample; assess liquidity, turnover, costs, and tracking-error sources.

Sources. Public bond ETF holdings and fact sheets; FINRA Fixed Income Data; instructor benchmark weights.

Deliverables. Benchmark table, cells, 10–20 bond sample, exposure gaps, tracking-error budget, and a 10–12 minute presentation.

Rubric (100 points). Benchmark diagnosis 20; cell design 25; selection 25; implementation tradeoffs 20; presentation and documentation 10.

2 Project 2: Rebalance Under Real-World Constraints

Scenario. Bring a drifting portfolio back within limits using no more than five trades.

Objectives. Diagnose active duration, sector, issuer, and liquidity exposures; choose trades; include lot sizes, bid-ask costs, and turnover.

Sources. Public ETF holdings; FINRA TRACE; issuer filings; instructor portfolio and bid-ask snapshot.

Deliverables. Before/after dashboard, trade list and rationale, transaction-cost estimate, rejected alternatives, and a 10–12 minute presentation.

Rubric (100 points). Diagnosis 25; trade design 25; constraints and costs 25; decision quality 15; presentation and reproducibility 10.

3 Project 3: Bullet, Barbell, or Ladder?

Scenario. Recommend a structure for a five-year income mandate facing uncertain curve changes and annual withdrawals.

Objectives. Construct duration-matched bullet, barbell, and ladder portfolios; compare cash-flow timing, convexity, reinvestment risk, liquidity, and behavior under steepening and flattening.

Sources. U.S. Treasury or Bank of Canada terms and yields; instructor withdrawal schedule and transaction-cost assumptions.

Deliverables. Three portfolio tables, cash-flow map, four curve scenarios, tradeoff scorecard, recommendation, and a 10–12 minute presentation.

Rubric (100 points). Portfolio construction 25; exposure matching 20; scenarios 25; recommendation 20; presentation and sources 10.

4 Project 4: Design an Investment-Grade Mandate

Scenario. Translate a nonprofit board's goals into an implementable bond portfolio policy and model allocation.

Objectives. Define return objective and risk limits; allocate across government and credit sectors; set duration, quality, issuer, liquidity, and turnover constraints; explain governance and rebalancing.

Sources. Public institutional investment policies; broad bond ETF fact sheets; FRED or Bank of Canada yields; instructor client facts.

Deliverables. One-page policy, strategic allocation, compliance dashboard, two stress tests, implementation plan, and a 10–12 minute presentation.

Rubric (100 points). Mandate translation 25; allocation logic 25; constraints 20; stress tests and governance 20; presentation and attribution 10.